Chapter 1

Points, Lines, Planes, and Angles

Unit 1 — Lines and Angles

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Point	A location with no size or shape
Line	Infinitely many points extending in two directions
Plane	Flat surface extending infinitely in all directions
Segment	Part of a line with two endpoints
Ray	Part of a line with one endpoint, extending in one direction
Collinear	Points on the same line
Coplanar	Points or lines on the same plane

Postulate 1-1: Unique Line

> Through any **two points**, there is exactly **one line**.

$$\text{Points } A \text{ and } B \Rightarrow \overleftrightarrow{AB} \text{ is unique}$$

Diagram: `A ———————— B` (only one line passes through both)

Postulate 1-2: Line Intersection

> If two distinct lines **intersect**, they intersect in exactly **one point**.

$$\ell_1 \cap \ell_2 = \{P\}$$

Note: Two lines either are parallel (never intersect), identical, or intersect at exactly one point.

Postulate 1-3: Plane Intersection

> If two distinct planes **intersect**, they intersect in exactly **one line**.

Example: The floor and a wall of a room intersect along one edge (a line).

Postulate 1-4: Unique Plane

> Through any **three noncollinear points** there is exactly **one plane**.

Key word: *Noncollinear* — the three points must not all lie on the same line.

Postulate 1-5: Ruler Postulate

> The points on a line can be paired one-to-one with the **real numbers**.
> The distance between points A and B is:

$$AB = |a - b|$$

where a and b are the coordinates of A and B.

Postulate 1-6: Segment Addition Postulate

> If B is between A and C, then:

$$AB + BC = AC$$

Example: If $AB = 5$ and $BC = 8$, then $AC = 13$.

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A —5— B —8— C
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Midpoint Theorem

> The **midpoint** M of segment $\overline{AB}$ divides it into two congruent segments:

$$AM = MB = \frac{AB}{2}$$

Coordinate Midpoint Formula:

$$M = \left(\frac{x_1 + x_2}{2}, \ \frac{y_1 + y_2}{2} \right)$$

Example — Midpoint

Find the midpoint of $A(2, 5)$ and $B(8, 1)$.

$$M = \left(\frac{2+8}{2},\ \frac{5+1}{2}\right) = (5,\ 3)$$

Check: Distance from A to M = Distance from M to B.

Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example: Distance from $A(1, 2)$ to $B(4, 6)$:

$$d = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25} = 5$$

Postulate 1-7: Protractor Postulate

> Rays from a point can be paired with real numbers from 0° to 180°.
> The measure of an angle is the absolute difference of its ray coordinates.

$$m\angle AOB = |r_1 - r_2|$$

Postulate 1-8: Angle Addition Postulate

> If D is in the interior of $\angle ABC$, then:

$$m\angle ABD + m\angle DBC = m\angle ABC$$

Example: If $m\angle ABD = 35^\circ$ and $m\angle DBC = 50^\circ$, then $m\angle ABC = 85^\circ$.

Angle Pairs: Definitions

Pair	Definition	Sum
Complementary	Two angles	$= 90^\circ$
Supplementary	Two angles	$= 180^\circ$
Linear Pair	Adjacent, supplementary angles	$= 180^\circ$
Vertical Angles	Opposite angles from two intersecting lines	Equal

Vertical Angles Theorem (Theorem 1-1)

> **Vertical angles are congruent.**

$$\angle 1 \cong \angle 3 \quad \text{and} \quad \angle 2 \cong \angle 4$$

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  \  /
2  \/  1
---/\---
3  /\  4
  /  \
```

If $\angle 1 = 70^\circ$, then $\angle 3 = 70^\circ$ and $\angle 2 = \angle 4 = 110^\circ$.

Supplement Theorem (Theorem 1-2)

> If two angles form a **linear pair**, they are **supplementary**.

$$m\angle 1 + m\angle 2 = 180^\circ$$

Complement Theorem (Theorem 1-3)

> If the noncommon sides of two adjacent angles form a **right angle**, the angles are **complementary**.

$$m\angle 1 + m\angle 2 = 90^\circ$$

Chapter 1 — Theorem Summary

#	Name	Statement
Post. 1-1	Unique Line	2 points $\to$ 1 line
Post. 1-2	Line Intersection	2 lines $\to$ 1 point
Post. 1-3	Plane Intersection	2 planes $\to$ 1 line
Post. 1-4	Unique Plane	3 noncollinear pts $\to$ 1 plane
Post. 1-6	Segment Addition	$AB + BC = AC$
Post. 1-8	Angle Addition	$m\angle ABD + m\angle DBC = m\angle ABC$
Thm. 1-1	Vertical Angles	Vertical \angles are $\cong$
Thm. 1-2	Supplement	Linear pair $\to$ supplementary
Thm. 1-3	Complement	Right angle $\to$ complementary pair